

Problem Statement ID : SIH25043

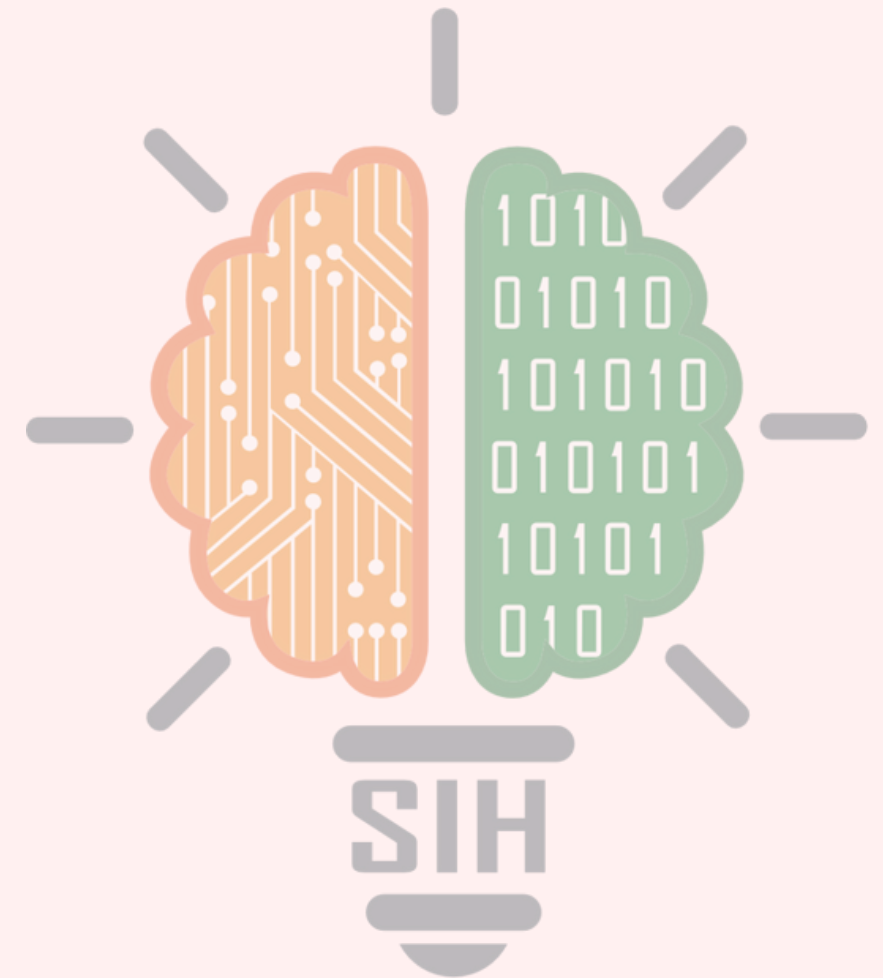
Problem Statement Title : Embedded Intelligent Microscopy System for Identification and Counting of Microscopic Marine Organisms

Theme : Smart Automation

PS Category : Hardware

Team ID:

Team Name : Team LimitLess



AquaNexus

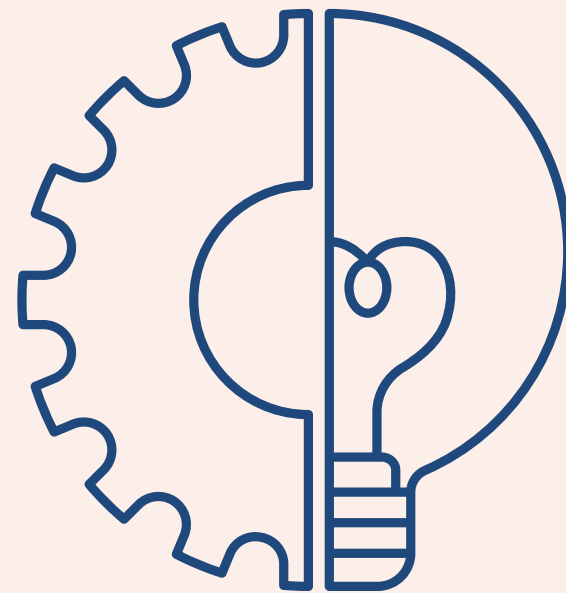
Core Capabilities

- **Self Improving AI:**
An **integrated LLM** (Gemini) acts as a **verification layer**, flagging potential misclassifications. This **feedback loop** allows for continuous, offline model improvement, **enhancing accuracy** over time.
- **Tamper Proof Decentralised Data:**
Utilises **Web3 technology** to create a **secure**, **shareable**, and **permanent log** of marine biodiversity data.

Scalability & Sustainability

- **Global Scalability** – Can be deployed across **7,500 km** of the **Indian coastline** and **scaled globally** without infrastructure changes.
- **Edge Design** – No dependency on high cost **cloud servers**, reducing operational costs.
- **Green Tech** – Designed for **minimal energy** consumption, making it **sustainable** and suitable for **long-term field deployment**.

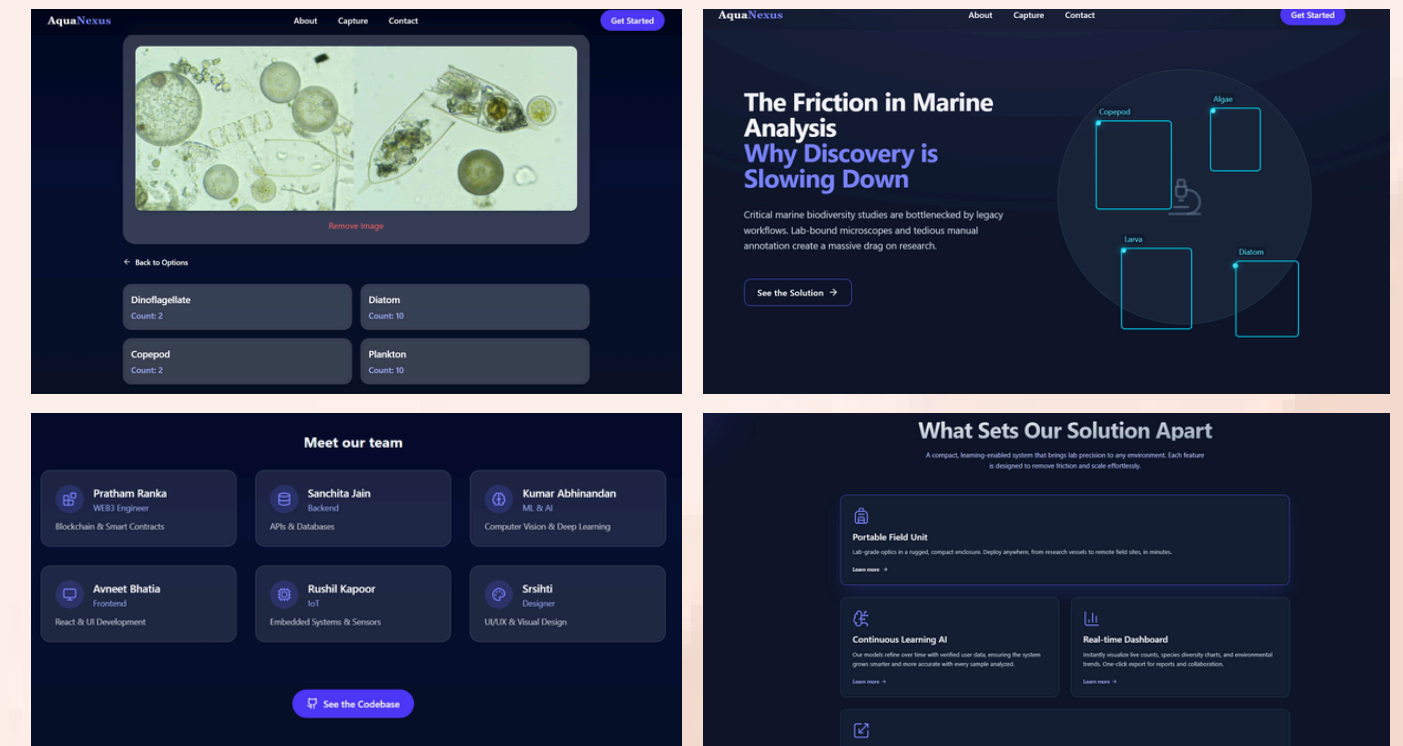
Our Innovation



Human Centred Design

- **Intuitive & Accessible:**
Simple plug-and-play design for fishermen, students, and scientists, no **technical expertise needed**
- **Offline Capability:**
Runs **100%** on-device with **Raspberry Pi + YOLOv8**, meaning it works even with **zero internet**, critical for **remote coasts**.

Deliverables (Live Link: aqua-nexus.vercel.app)

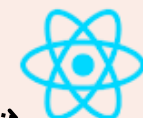
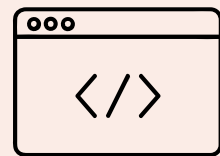


TECHNICAL APPROACH

Our AI microscope is **plug and play**. A USB camera connected to a microscope streams to a **Raspberry Pi**, where **YOLOv8** **detects** and **counts** plankton. Our **model** tells the name of the **Species**, and **Gemini** verifies if the classification is correct. Results appear on **screen**, and every **validated record** is pinned to **IPFS** for **secure, tamper proof data**.

TECH STACK

Frontend



FastAPI

Backend



R
ReactJS

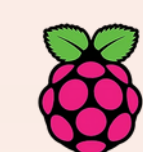
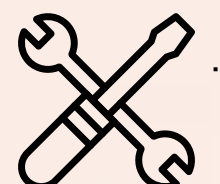
TS

AI/ML



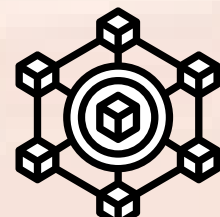
OpenCV

IOT



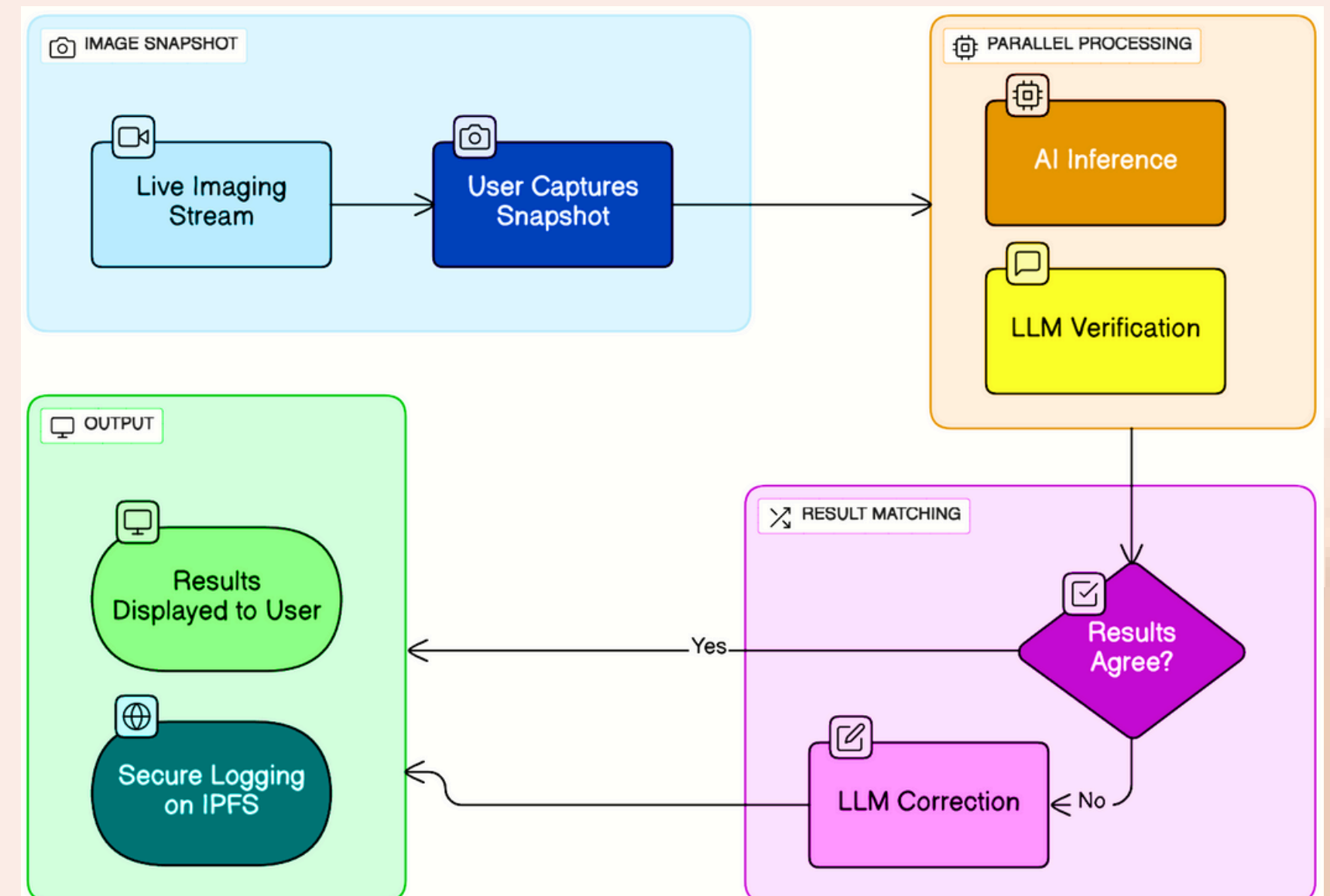
ARDUINO

WEB 3.0

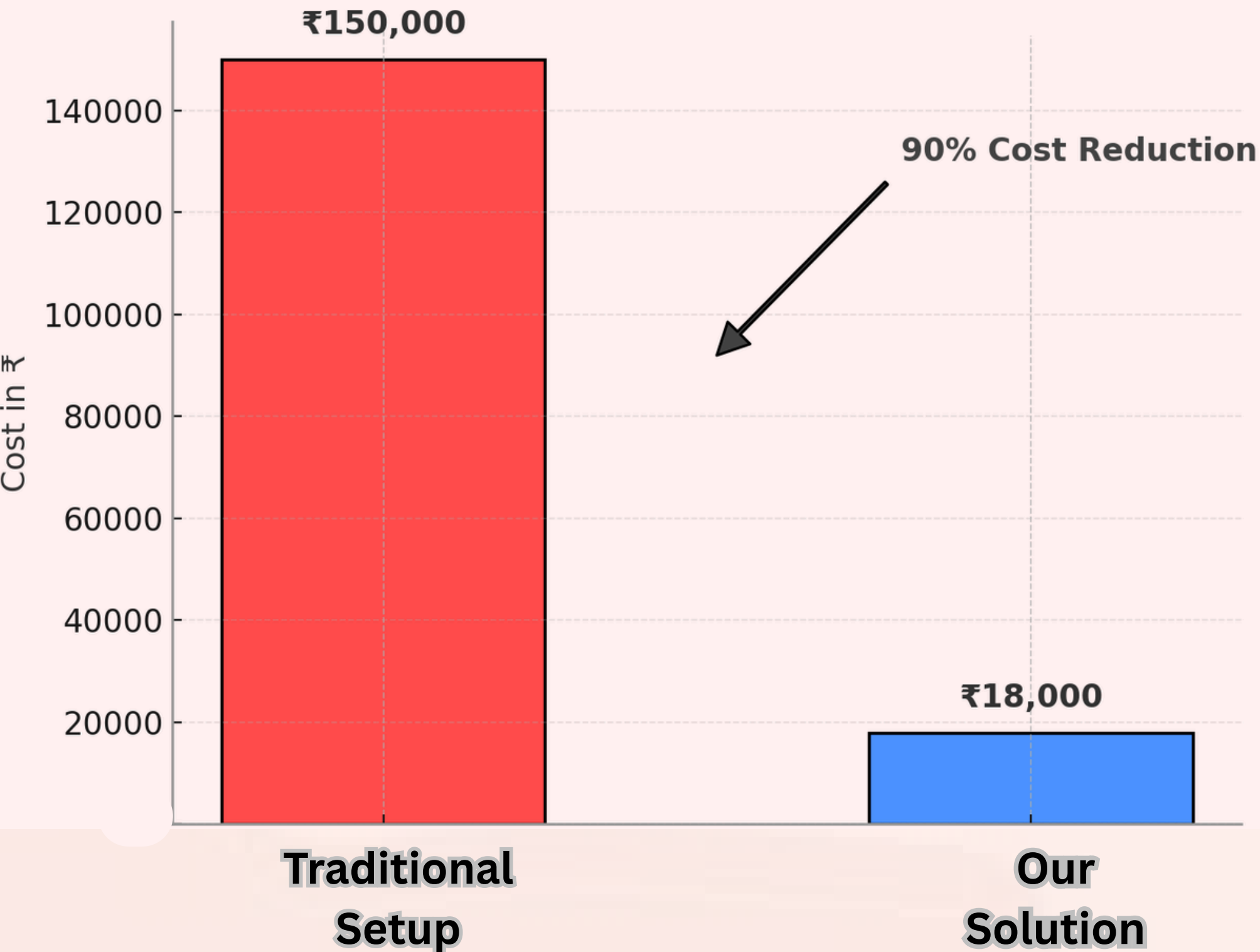


IPFS

SYSTEM ARCHITECTURE



FEASIBILITY AND VIABILITY



◆ Technically Superior & Commercially Defensible

Our **integrated system** (hardware + dual-AI) delivers analysis in under **7 seconds** with a **90% cost reduction** over traditional methods.

◆ Proven Market Need & Clear Adoption Path

From a **48-hour lab** wait to an **instant field result**, our tool's value is clear. We'll leverage fisheries co-ops for rapid adoption and impact on **India's \$30B marine economy**.

◆ Exponential Value via a Future-Ready Platform

Our platform will evolve beyond data collection, using **AI, Vision Transformer** and **new sensors** to generate **revenue** from **biodiversity audits** and **carbon credits**, creating a strategic national asset.

IMPACT AND BENEFITS



Driving India's National & Global Vision

Promotes **AI** for **Societal Good** by enabling real time marine biodiversity assessment, advancing **India's Blue Economy Vision 2030**.



Supports Global & National Sustainability Goals

Contributes to **SDG 14** (Life Below Water) and **SDG 13** (Climate Action) through **science-based conservation** and **climate resilience strategies**.



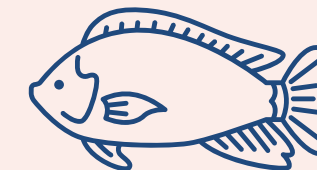
Decentralized & Transparent Data Sharing (IPFS)

For open data sharing, breaking silos, **ensuring transparency** and trust, while positioning India as a global pioneer in **AI driven marine innovation** aligned with **Atmanirbhar Bharat**.



Sustainable Governance & Economic Security

By predicting **algal blooms**, we prevent costly disasters like the **₹41 crore Kerala fish kill**. This safeguards livelihoods and provides vital data for **Digital India missions**.



Empowering Coastal Communities & Safeguarding Fisheries

Empowers **fishing communities** with **real time alerts**, reducing **economic losses** and enhancing **sustainable fisheries** management in India's **3rd largest** fish producing economy.

RESEARCH & REFERENCES

- **Indian AI Mission & Policy:** <https://www.psa.gov.in/ai-mission>
- **Digital Ocean Indian Data:** <https://www.pib.gov.in/PressReleaseDetailm.aspx?PRID=1684418#:~:text=Highlighting%20the%20benefits%20of%20Digital,%E2%80%9D>
- **Indian Blue Economy & Tech:** <https://kpmg.com/in/en/blogs/2024/10/unlocking-indias-blue-economy-a-pathway-to-sustainable-growth-and-prosperity.html#:~:text=4>
- **Vision Transformer Research Paper:** <https://www.mdpi.com/2313-433X/11/7/219>
- **SDG Alignment:**
 - **SDG 14:** <https://sdgs.un.org/>
 - **SDG 13:** <https://sdgs.un.org/>
- **Technical Foundations:** <https://shorturl.at/Fgk2b>
- **Kerala Periyar Fish Kill:** <https://www.thehindu.com/news/cities/Kochi/peoples-report-puts-periyar-fish-kill-loss-at-over-41-crore/article68445672.ece>